Library Management System

Members

Member_ID (PK)	Member_Name	City	Age	Email
M101	Ishaan Shah	Ahmedabad	21	ishaan@gmail.com
M102	Meera Patel	Mumbai	24	meera@gmail.com
M103	Rohan Mehta	Ahmedabad	22	NULL
M104	Kavya Joshi	Pune	27	kavya@gmail.com
M105	Arjun Shah	Surat	23	arjun@gmail.com
M106	Nisha Patel	Vadodara	26	NULL

Books

Book_ID (PK)	Book_Title	Category	Author	Price
B101	Database Systems	Technology	Korth	850
B102	Python Basics	Programming	Sarkar	700
B103	Web Development	Technology	Duckett	900
B104	Data Science Essentials	Data Science	Wickham	1200
B105	Cyber Security	Security	Stallings	1100
B106	Machine Learning	Data Science	Mitchell	1350

Borrowings

Borrow_ID (PK)	Member_ID (FK)	Book_ID (FK)	Borrow_Date	Days_Borrowed	Fine
BR001	M101	B101	2026-01-05	7	0
BR002	M102	B102	2026-01-08	12	50
BR003	M103	B104	2026-01-10	5	0
BR004	M104	B103	2026-01-12	15	100
BR005	M105	B105	2026-01-15	9	20
BR006	M106	B106	2026-01-18	18	150
BR007	M101	B104	2026-02-01	11	40
BR008	M103	B105	2026-02-04	14	60
BR009	M102	B103	2026-02-07	6	0
BR010	M104	B102	2026-02-09	21	200
BR011	M105	B106	2026-02-12	8	0
BR012	M106	B101	2026-02-15	13	30

Query Questions

1. Display the names and cities of members whose names start with 'A' or 'M'.

2. Display members whose names contain the letter 'a' and whose age is between 22 and 26.

3. Display members who belong to Ahmedabad, Pune, or Vadodara and whose age is not 22.

4. Display books whose titles contain the word 'Data' or whose category is Technology.

5. Display borrowings where Days_Borrowed is between 8 and 15 and Fine is between 20 and 100.

6. Display borrowings for books B101, B104, or B106 where Days_Borrowed is greater than 10.

7. Display members who do not live in Mumbai or Surat and whose age is greater than 23.

8. Display books whose price is between ₹800 and ₹1,200 but whose category is not Security.

9. Display borrowings ordered by Fine from highest to lowest and then by Days_Borrowed from highest to lowest.

10. Display members ordered first by City in ascending order and then by Age in descending order.

11. Display the different cities represented by members in alphabetical order.

12. Display the different fine amounts in descending order.

13. Display the five borrowings with the greatest number of Days_Borrowed.

14. Display the three least expensive books.

15. Find the total fine collected from all borrowings.

16. Find the average number of Days_Borrowed for borrowings where Fine is greater than 0.

17. Find the total number of borrowing records.

18. Count the members whose email addresses are available.

19. Find the highest Fine charged.

20. Find the lowest number of Days_Borrowed recorded.

21. Find the total, average, maximum, and minimum Fine values.

22. Find the total fine for each Book_ID and display only books whose total fine is greater than ₹50.

23. Find the average number of Days_Borrowed for each Book_ID and display only books whose average is greater than 10 days.

24. Find the number of times each Book_ID was borrowed and display only books borrowed at least twice.

25. Find the highest Fine for each Book_ID and display only books whose highest fine is greater than ₹50.

26. Find the lowest Fine for each Book_ID and display only books whose lowest fine is greater than 0.

27. Find the total fine for each Member_ID and display only members whose total fine is greater than ₹50.

28. Find the average number of Days_Borrowed for each Member_ID and display only members whose average borrowing period is greater than 10 days.

29. Find the number of books borrowed by each Member_ID and display only members who borrowed at least 2 books.

30. Find the total fine for each Member_ID considering only borrowings with Days_Borrowed greater than 10, and display groups whose total fine is at least ₹50.

31. Find the average Fine for each Book_ID considering only borrowings with Days_Borrowed at least 10, and display groups whose average fine is greater than ₹40.

32. Find the number of borrowings for each Book_ID considering only records with Fine greater than 0, and display books having at least 1 such record.

33. Find the maximum Days_Borrowed for each Member_ID and display members whose maximum borrowing period is greater than 15 days.

34. Find the minimum Days_Borrowed for each Member_ID and display members whose minimum borrowing period is at least 8 days.

35. Find the total fine for each Book_ID, arrange the groups from highest total fine to lowest, and display only the first 3 groups.

36. Find the average borrowing period for each Member_ID, arrange the groups from highest average to lowest, and display the top 3 groups.

37. Find the number of borrowings for each Member_ID, arrange the groups from highest count to lowest, and display the first 3 groups.

38. Display members whose email address is missing.

39. Display members whose email address is available and whose city is not Surat.

40. Change the age of member M103 from 22 to 23.

41. Increase the price of all books costing less than ₹900 by ₹100.

42. Increase the Fine of all borrowings where Days_Borrowed is greater than 15 by ₹25.

43. Change the Fine of borrowing BR005 to ₹30.

44. Delete borrowing record BR009.

45. Delete borrowings having a Fine of 0.

46. Create a view containing Borrow_ID, Member_ID, Book_ID, Days_Borrowed, and Fine for borrowings with Days_Borrowed of at least 12.

47. Display records from the view where Fine is at least ₹50.

48. Create a view containing Book_ID, Book_Title, Category, and Price for books whose price is at least ₹1,000.

49. Display records from the book view belonging to the Data Science or Security category.

50. Create a view containing Member_ID, Member_Name, City, and Age for members whose age is at least 23.

51. Display records from the member view for members from Ahmedabad, Pune, or Vadodara.